

1 • Input

Molecular formula + literature
peak lists (IR cm^{-1} , ^1H , ^{13}C)

blind peak lists (not traces)



2 • Generation

LLM proposes ranked
candidate structures

true enters pool 65/194 (34%)



3 • Verification

Forward-verify / re-rank
candidates against spectra

selects true 58/65 (89%) when present

top-1 = generation recall $\times$ verification precision|recall

$$30\% \approx 34\% \times 89\%$$

65 recalled (34%) forward-verified top-1 58/194 = 30%; precision|recall 58/65 = 89%



verified

mis-ranked

never proposed

No re-ranking repairs the 129 never proposed

Verification alone: 28% $\rightarrow$ 30% (diagnostic, not an accuracy advance)